

VIO Evaluation Report

Generated: 2026-04-15 17:42:24 | Log: logs/survey-mission/

1. Summary

Trajectory Info		Position Error (ATE)	
GT Length (m)	1566.4	RMSE	3.778
VIO Length (m)	1578.1	Mean	3.550
Duration (s)	328.3	Median	3.598
Yaw Alignment (deg)	-1.8	Std	1.293
GT Samples	8825	Min	0.139
VIO Samples	1398	Max	8.541
		Drift %	0.241

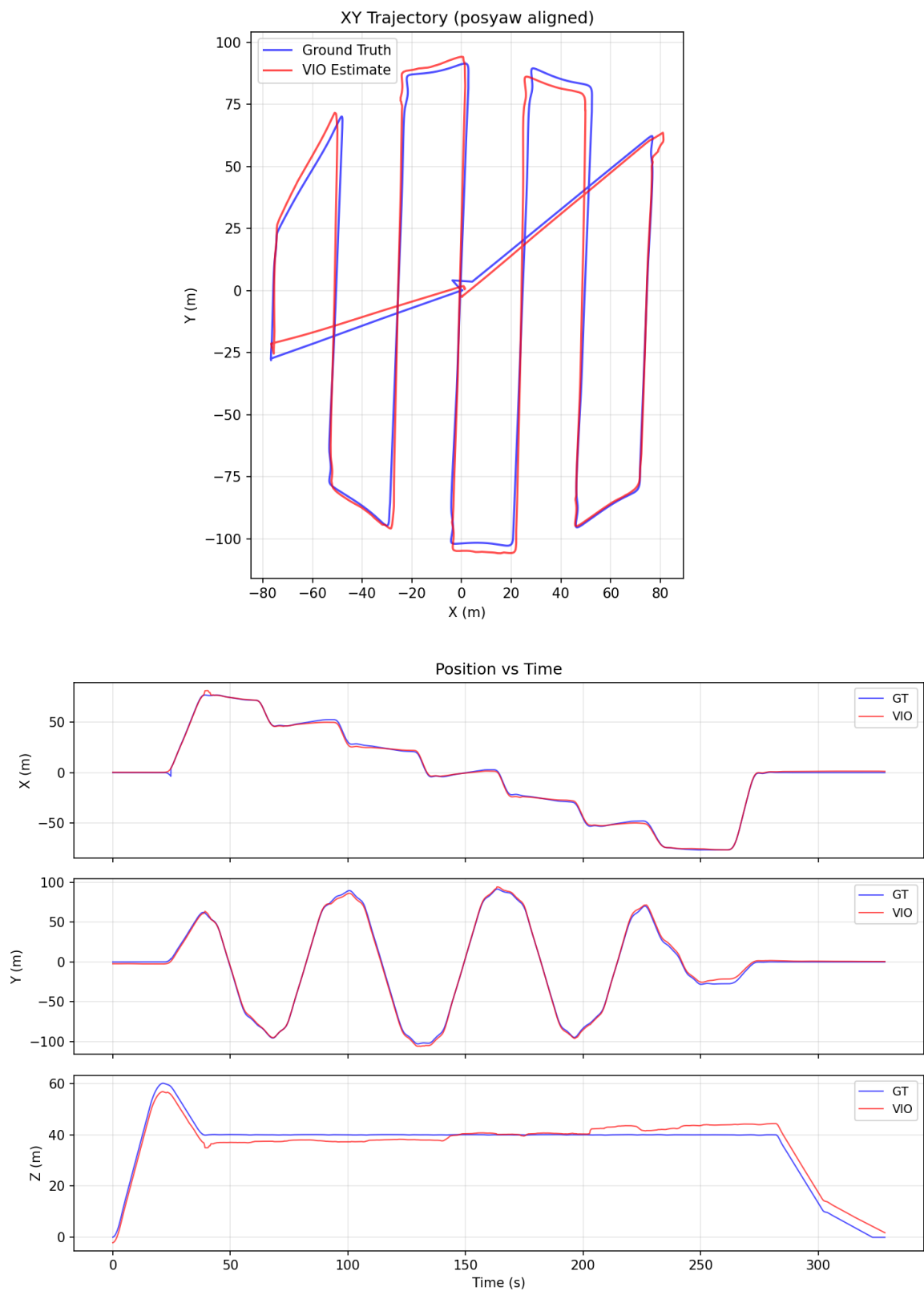
Velocity Error		Relative Pose Error (median m)	
Speed RMSE	0.164	8m (1335 pts)	0.329
Speed Mean Abs	0.114	16m (1291 pts)	0.552
3D Vel RMSE	0.421	24m (1271 pts)	0.754
3D Vel Mean	0.198	32m (1251 pts)	0.959
Vx RMSE	0.381	40m (1234 pts)	1.168
Vy RMSE	0.166		
Vz RMSE	0.064		
Max GT Speed	10.57		

Overall Assessment

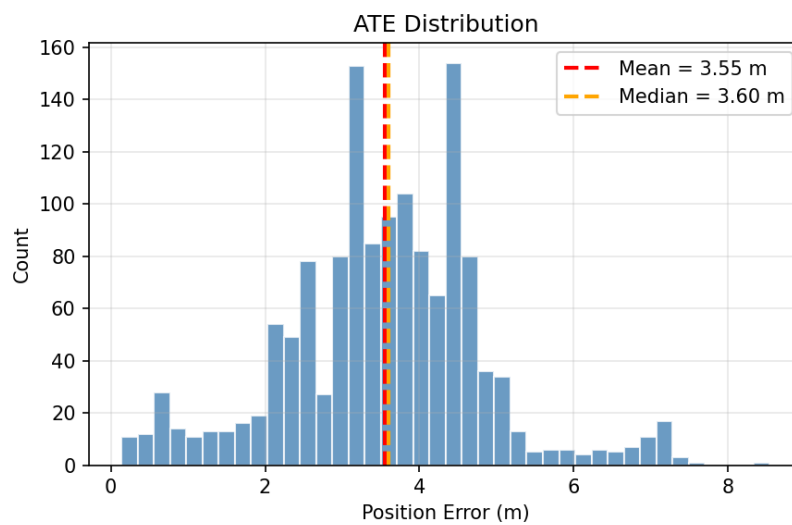
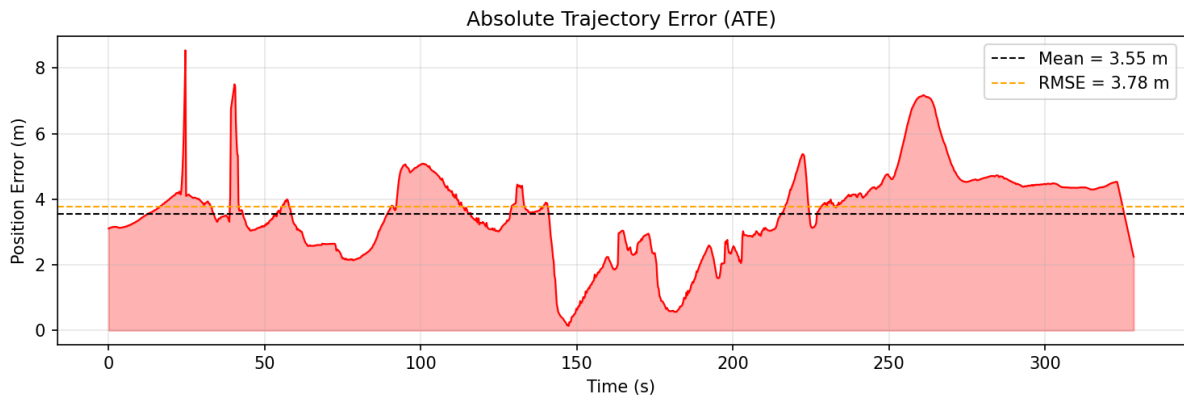
Metric	Value	Grade
Position Drift	0.241%	Excellent
ATE RMSE	3.778 m	—
Speed RMSE	0.164 m/s	Excellent

Grading: Drift <0.5%=Excellent, <1%=Good, <2%=Acceptable. Speed RMSE <0.2=Excellent, <0.5=Good, <1.0=Acceptable.

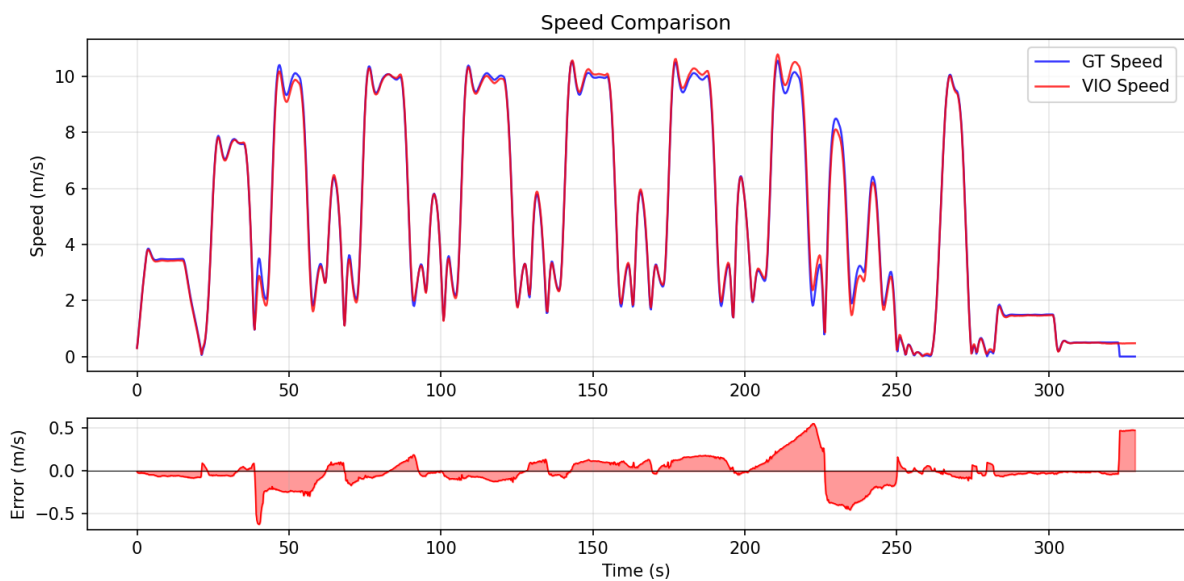
2. Trajectory Comparison



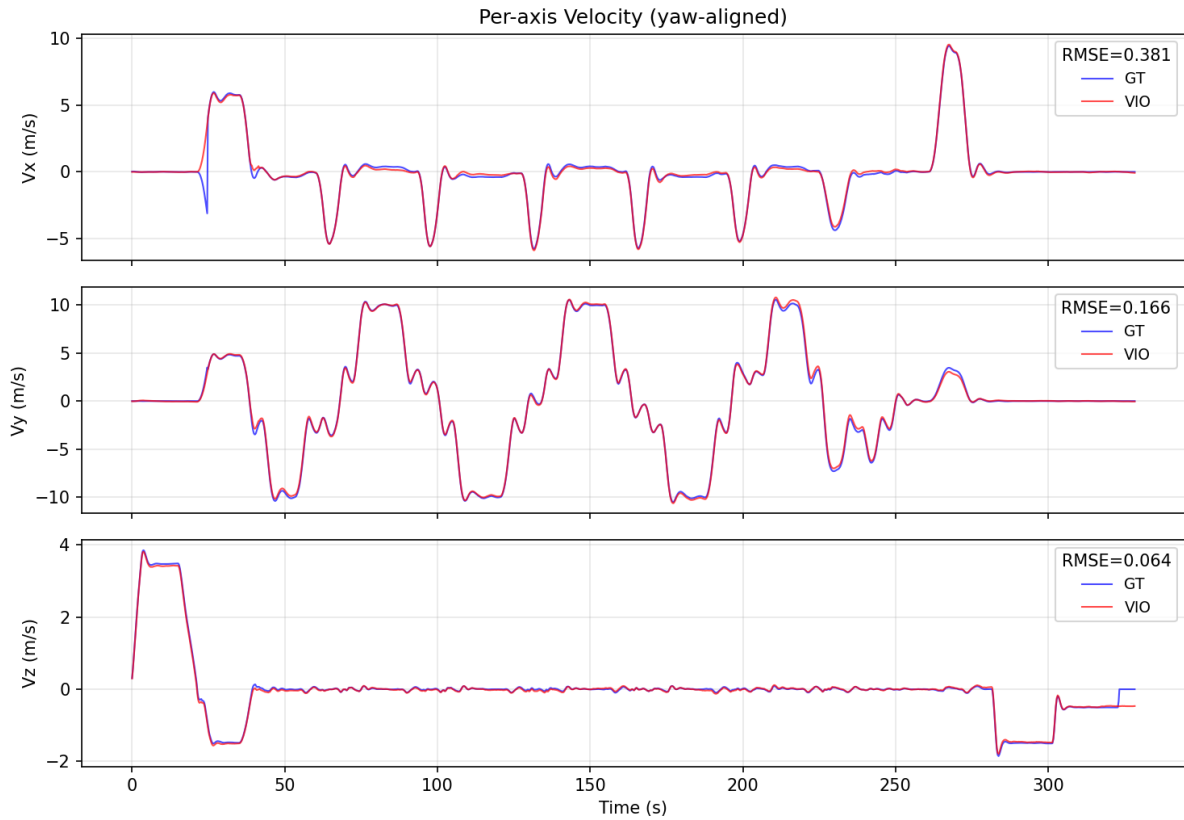
3. Absolute Trajectory Error



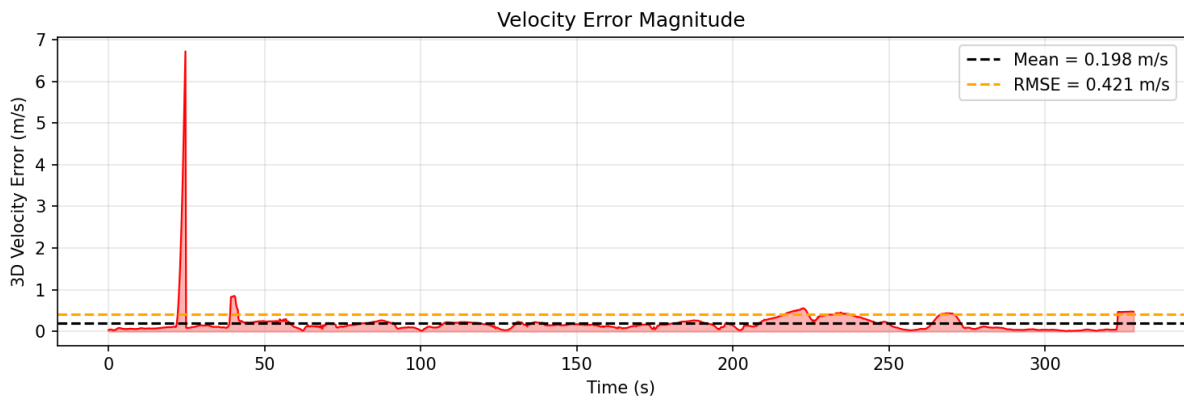
4. Speed Comparison



5. Per-axis Velocity



6. Velocity Error



6. Overall Assessment

Metric	Value	Grade
Position Drift	0.241%	Excellent
ATE RMSE	3.778 m	
Speed RMSE	0.164 m/s	Excellent

Grading: Position drift <0.5% = Excellent, <1% = Good, <2% = Acceptable. Speed RMSE <0.2 m/s = Excellent, <0.5 m/s = Good, <1.0 m/s = Acceptable.